

Notes 3 — Linux GUI, CLI, and Commands

1. What is a graphical user interface (GUI)?

A **graphical user interface (GUI)** is a set of programs that allows a user to interact with the computer system via icons, windows, and various other visual elements.

Example

- Clicking on a folder to open it
- Using a web browser
- Moving files by dragging them into another folder

2. What is a desktop environment?

A **desktop environment** is an implementation of the desktop metaphor made of a bunch of programs running on top of a computer operating system which share a graphical shell. It includes things like windows, menus, icons, and applications.

Example

- GNOME
- KDE

3. What is the command line interface (CLI)?

A **command line interface (CLI)** is a way to communicate with the computer by typing commands. Instead of clicking on icons and menus, you type a command and the computer gives you a result.

Example

```
bash date
```

This command displays the current date and time.

4. How do I access the command line interface (CLI)?

There are several ways to access the command line in Linux. You can open a **terminal emulator** from the graphical desktop. You can also use a **linux console** because it is a direct interface to the linux system.

Example `ls -l /home/user`

5. What is a virtual console?

A **virtual console** is a terminal session that runs in linux system memory.

6. What is a terminal emulator?

A **terminal emulator** is a program that gives you access to the linux CLI when using the GUI.

Examples

- GNOME Terminal
- Konsole
- Kitty The terminal emulator lets you type commands and see the results without leaving the graphical desktop.

7. What is bash?

Bash stands for **Bourne Again Shell**. It is a shell that allows users to interact with the Linux operating system by typing commands. Bash can also be used to run scripts and automate tasks.

Example `bash echo Hello` This tells Bash to display the word **Hello**.

8. What is the shell prompt?

The **shell prompt** is the text shown in the terminal that tells you the shell is ready for you to enter a command.

Example `bash ->` When you see the prompt, you can type a command and press Enter.

Linux Commands

clear

Definition:

The **clear** command clears the text from the terminal screen.

Usage

It is useful when the terminal has become cluttered and you want a clean screen.

Example

```
bash clear
```

echo

Definition

The **echo** command displays text or other information on the screen.

Usage

It can be used to print messages or display the value of something.

Example

```
bash echo Hello World
```

- Output Hello World

date

Definition

The **date** command displays the current date and time.

Usage

It can be used when you need to quickly check the system's date and time.

Example

```
bash date
```

free

Definition

The **free** command displays information about the computer's memory.

Usage

It can be used to check how much memory is being used and how much is available.

Example

```
bash free
```

uname

Definition

The **uname** command displays information about the system.

Usage

It can be used to find information about the Linux system and kernel.

Example

```
bash uname -a
```

The **-a** option displays more information about the system.

history

Definition

The **history** command shows commands that were previously entered into the terminal.

Usage

It is useful when you want to find a command you used earlier without typing it again.

Example

bash history

man

Definition

The **man** command displays the manual page for a command.

Usage

It is useful for learning how a command works, including its options and how to use it.

Example

bash man echo This opens the manual page for the **echo** command.

tldr

Definition

tldr provides shorter and easier-to-read examples for Linux commands.

Usage

It is useful when the regular **man** page has too much information and you just want to see common examples.

Example

bash tldr echo This gives examples of how to use the **echo** command.

cheat

Definition

The **cheat** command is used to display cheat sheets for commands.

Usage

It can be useful when you want to quickly look up how to use a command.

Example

bash cheat echo This displays information and examples for using **echo**.

hostname

Definition

The **hostname** command displays the name of the computer.

Usage

It can be used to find the name assigned to the Linux computer.

Example

```
bash hostname
```

df

Definition

The **df** command shows information about the amount of disk space available on file systems.

Usage

It can be used to check how much storage space is being used and how much is still available.

Example

```
bash df -h
```

du

Definition

The **du** command shows how much disk space files and directories are using.

Usage

It can help you find out which files or folders are taking up storage space.

Example

```
bash du -h
```

figlet

Definition

The **figlet** command displays text in large letters made from characters.

Usage

It can be used to make text stand out in the terminal.

Example

```
bash figlet Linux
```

This displays the word **Linux** using large ASCII-style letters.